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SHORT COMMUNICATION

Prevalence of Parkinson's disease: a population-based study in Portugal

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Background and purpose: Portugal has been identified as one of the countries with a high prevalence of LRRK2-G2019S, considered to be the most frequent known cause of familial and sporadic Parkinson's disease (PD). The aim of this study was to evaluate the prevalence of PD in Portugal using a door-to-door methodology.

Methods: A cross-sectional study was conducted in the Portuguese community-dwelling population; that is, elderly people living in the community on their own, aged ≥50 years and resident in mainland Portugal, in two phases: (i) a questionnaire was applied to screen potential cases of PD; and (ii) screened cases were evaluated by an expert in PD to confirm diagnosis.

Results: The adjusted prevalence of PD for the Portuguese community-dwelling population aged ≥50 years was 0.24%. The estimated total number of cases of PD for the Portuguese population is 180/100 000 inhabitants.

Conclusions: The results of this study show that a geographical region with a high frequency of a causal mutation for PD does not automatically imply a high prevalence of patients with PD.

Introduction

Mutation in the leucine-rich repeat kinase 2 (LRRK2) gene is considered to be the most frequent known cause of familial and sporadic Parkinson's disease (PD), and G2019S (c.G6055A) has been shown to be the most frequent known LRRK2 mutation. However, it is currently unknown whether the frequency of this mutation in the population increases the prevalence of PD [1].

Although the mutation has a worldwide distribution, Portugal has been identified as one of the countries with a high frequency of LRRK2-G2019S, with prevalence between 3.7% and 4.9% for sporadic cases and 9.1% and 16.1% for familial cases [2].

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Methods

A community-dwelling population-based; that is, elderly people living in the community on their own, cross-sectional study was conducted in the Portuguese population aged ≥50 years resident in mainland Portugal. The study was approved by the Ethics Committee of the Faculty of Medicine of the University of Lisbon. The study was conducted in two phases: (i) a validated translated questionnaire was applied in person by a trained interviewer, after oral consent was given, to screen for cases of PD; and (ii) screened cases were contacted to schedule an appointment with an expert in PD, after giving written consent, to confirm diagnosis. The questionnaire included demographic characterization, a presumption scale and information regarding previous diagnosis of PD and treatment. The presumption scale consisted of a set of nine questions (rated between 6 and 9 points) and a presumption diagnosis

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of PD was assumed if the total score was >42 points. This scale was translated into and adapted to the Portuguese language and subjected to a validation process for the purpose of this study. The construct validation was performed by applying the scale to 10 patients with PD and 10 negative PD cases. For external validation, the presumption scale was applied to 50 negative PD cases.

Positive screening cases had to fulfil at least one of the following conditions: presumption scale score >42 points [3], previous diagnosis of PD and/or presence of antiparkinson medication.

The sample was stratified into Portugal's Nomenclature of Territorial Units for Statistics II and collected through a random walking method, where the interviewer followed a random route to select households, e.g. take the first road right, interview the second house on the left, continue down the road, interview the third household on the right, etc. At each household, interviewers were allowed to apply the questionnaire to just one member who fitted the population criteria. Nursing homes and senior citizens' residences were excluded.

Statistics

The sample size was calculated to determine the prevalence of PD in the Portuguese population ≥50 years old. An estimated prevalence of 0.5% was assumed and, with a margin of error of ±0.2% and a confidence interval (CI) of 95%, an estimated sample size of 5000 individuals was defined.

Statistical analysis estimated the prevalence of PD, age-standardized to the European population structure, and the total number of cases of PD. 95% CIs were established for the estimates.

Results

A total of 5048 people were screened for PD. The sample age varied between 50 years (lower limit defined) and 95 years [55.6% female; mean age 65.1 ± 10.0 years (SD)]. A total of 0.3% (n = 14) had a previous PD diagnosis and 0.3% (n = 14) were taking antiparkinson medication. A total of 0.4% (n = 22) scored >42 points on the presumption scale.

A total of 22 people were screened for PD, 12 of them confirmed by the neurologist from the expert panel (Fig. 1).

From these data, the adjusted prevalence of PD for the Portuguese population aged ≥50 years was 0.24% (95% CI, 0.043–0.437%). The estimated prevalence of PD was higher for men (P = 0.033) and increased significantly with age (P < 0.001) (Figure S1).

The total number of cases of PD estimated for the mainland Portuguese population ≥50 years of age was 240/100 000 inhabitants. When extrapolating to the Portuguese population, the estimated total number of cases of PD was 180/100 000 inhabitants (95% CI, 30–327/100 000). The presumption scale used in the screening phase to identify patients with PD had a high specificity and low sensitivity (Table 1) demonstrating that, in the absence of a previous PD diagnosis and/or presence of PD medication, this screening tool has a low power to detect potential PD.

Discussion

The prevalence of PD was lower than expected for a country with a very high prevalence of LRRK2-G2019S. Compared with prevalence data from other European countries obtained with similar door-to-door methodology, these results show a lower number of

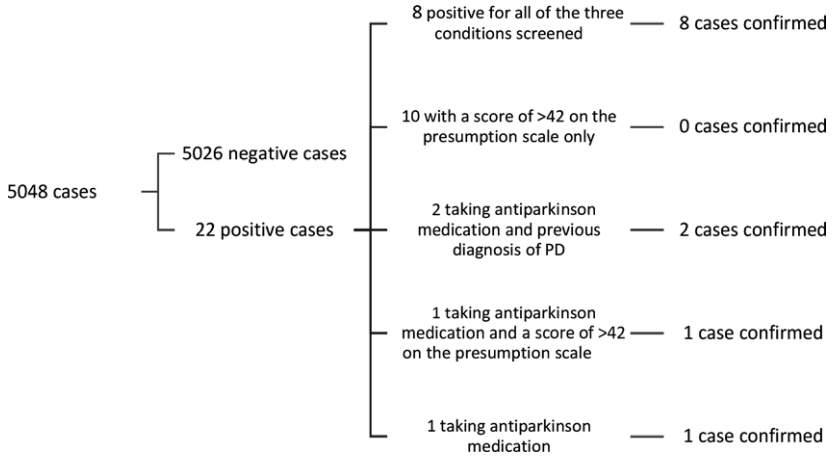


Figure 1 Cases of Parkinson's disease (PD) identified and confirmed.

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Table 1 Sensitivity and specificity analysis of the scale for presumption diagnosis of Parkinson's disease

Parameter	
Sensitivity	75.0% (44.9–93.1)
Specificity	99.7% (99.7–99.8)
Positive predictive value	40.9% (24.5–50.8)
Negative predictive value	99.9% (99.9–100.0)

Data are given as % (95% confidence interval).

cases of PD per 100 000 inhabitants, specifically when compared with Northern European countries such as Germany (713/100 000) [4], the Netherlands (1400/100 000) [5] or France (328/100 000) [6], or with Southern European countries such as Italy (257/100 000) [7] or Spain (1280/100 000) [3]. However, as the estimated number of patients with PD in Portugal is based on a very small number of identified patients, the estimate has a wide confidence interval, making the upper limit similar to the results from studies in France and Italy.

Methodological and design limitations offer some hypotheses for the low prevalence estimates. The first limitation concerns the required sample size that was calculated assuming a prevalence of 0.5%, which was above the actual prevalence obtained. The sample was therefore underpowered to detect cases of PD. The second limitation is that a self-reported questionnaire was applied for screening cases of PD. The last limitation concerns the sample size design, which excluded nursing homes and senior citizens' residences, meaning that the more severe cases were missed. It is not uncommon to observe large variations in the prevalence of PD in results from door-to-door studies compared with studies that include nursing homes. Therefore, further studies including institutionalized patients are needed to achieve a more realistic estimate of the prevalence of PD and total number of cases. The results of this study show that a geographical region with a high frequency of a causal mutation for PD does not automatically imply a high prevalence of patients with PD.

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Disclosure of conflicts of interest

J.J.F. held consultancies with Ipsen, GlaxoSmithKline, Novartis, Teva, Lundbeck, Solvay, Abbott, BIAL, Merck-Serono and Merz, and received grants from GlaxoSmithKline, Grunenthal, Teva and Fundação MSD. All other authors declare no financial or other conflicts of interest.

Supporting Information

Additional Supporting Information may be found in the online version of this article:

Figure S1. Parkinson's disease prevalence by gender and group age (95% confidence interval).

References

1. Correia Guedes L, Ferreira JJ, Rosa MM, Coelho M, Bonifati V, Sampaio C. Worldwide frequency of G2019S LRRK2 mutation in Parkinson's disease: a systematic review. *Parkinsonism Relat Disord* 2010; **16:** 237–242.
2. Ferreira JJ, Guedes LC, Rosa MM, *et al*. High prevalence of LRRK2 mutations in familial and sporadic Parkinson's disease in Portugal. *Mov Disord* 2007; **22:** 1194–1201.
3. Claveria LE, Duarte J, Sevillano MD, *et al*. Prevalence of Parkinson's disease in Cantalejo, Spain: a door-to-door survey. *Mov Disord* 2002; **17:** 242–249.
4. Trenkwalder C, Schwarz J, Gebhard J, *et al*. Starnberg trial on epidemiology of Parkinsonism and hypertension in the elderly. Prevalence of Parkinson's disease and related disorders assessed by a door-to-door survey of inhabitants older than 65 years. *Arch Neurol* 1995; **52:** 1017–1022.
5. de Rijk MC, Breteler MM, Graveland GA, *et al*. Prevalence of Parkinson's disease in the elderly: the Rotterdam Study. *Neurology* 1995; **45:** 2143–2146.
6. Munoz M, Boutros-Toni F, Preux PM, *et al*. Prevalence of neurological disorders in Haute-Vienne department (Limousin region-France). *Neuroepidemiology* 1995; **14:** 193–198.
7. Morgante L, Rocca WA, Di Rosa AE, *et al*. Prevalence of Parkinson's disease and other types of parkinsonism: a door-to-door survey in three Sicilian municipalities. The Sicilian Neuro-Epidemiologic Study (SNES) Group. *Neurology* 1992; **42:** 1901–1907.

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